TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 8

5-MINUTE REVISION

Measures of Association

From the chapter's own revision points, formulae and questions.

BOOK-ONLY EDITION · Verified against the textbook

Selected from Chapter 8 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Measures of association → describe relationships between variables.
- Linear relationship → relationship that can be represented by a straight line.
- Monotonic relationship → one variable consistently increases or consistently decreases as the other increases, but not necessarily at a constant rate.
- Scatter diagram → graphical method for examining association between two variables.
- Positive correlation → both variables tend to move in the same direction.

Quick revision · 2

- Negative correlation → variables tend to move in opposite directions.
- No correlation → no systematic linear association.
- Simple correlation → relationship between two variables.
- Multiple correlation → relationship involving more than two variables.
- Linear correlation → relationship represented by a straight line.

Quick revision · 3

- Non-linear correlation → relationship represented by a curved pattern.
- Partial correlation → relationship between two variables after controlling other variables.
- Perfect correlation → all points lie on a straight line.
- Pearson's correlation → measures linear association.
- Pearson's r ranges from -1 to $+1$.

Quick revision · 4

- $r = +1$ → perfect positive correlation.
- $r = -1$ → perfect negative correlation.
- $r = 0$ → no linear correlation.
- Covariance → measures joint linear variability.
- Covariance depends on the units of measurement.

Quick revision · 5

- Pearson's correlation is unit-free.
- Pearson's correlation is sensitive to outliers.
- Pearson's correlation summarises linear association; a nonlinear pattern can make it misleading as a general measure of association.
- Spearman's ρ → rank-based measure of monotonic association.
- Spearman's correlation is useful for ordinal data and non-linear monotonic relationships.

Quick revision · 6

- Spearman's correlation ranges from -1 to $+1$.
- No tied ranks → use the basic Spearman formula.
- Tied ranks → use average ranks and correction terms.
- Kendall's coefficient of concordance → measures agreement among multiple rankings.
- Correlation matrix → table of pairwise correlation coefficients.

Quick revision · 7

- Diagonal elements of a correlation matrix → 1.
- Correlation matrix → symmetrical about the diagonal.
- Correlogram → graphical representation of a correlation matrix.
- Strong positive correlation → coefficient close to $+1$.
- Strong negative correlation → coefficient close to -1.

Quick revision · 8

- Correlation measures association, not necessarily causation.

Important formulae · 1

Covariance:

$$cov(x,y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Pearson's correlation coefficient:

$$r = \frac{cov(x,y)}{sd(x)sd(y)}$$

Pearson's correlation coefficient using deviations:

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Important formulae · 2

Spearman's rank correlation coefficient without tied ranks:

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$$

where d_i is the difference between the ranks of the ith pair of observations.

Spearman's rank correlation coefficient with tied ranks:

$$\rho = \frac{2S_0 - T_x - T_y - \sum d_i^2}{2\sqrt{(S_0 - T_x)(S_0 - T_y)}}, \quad S_0 = \frac{n(n^2-1)}{12}$$

Correction for tied ranks in X:

$$T_x = \frac{1}{12}\sum_{i=1}^{s} m_i(m_i^2 - 1)$$

Important formulae · 3

Correction for tied ranks in Y:

$$T_y = \frac{1}{12}\sum_{i=1}^{s'} w_i(w_i^2 - 1)$$

Kendall's coefficient of concordance:

$$W = \frac{12\left[\sum_{i=1}^{n} R_i^2 - \frac{\left(\sum_{i=1}^{n} R_i\right)^2}{n}\right]}{k^2 n(n^2-1)}$$

where R_i is the sum of ranks obtained by the ith object and k is the number of judges.

Self-check from the chapter

1. The strength and direction of linear association are measured by Pearson's _____ coefficient. **[Correlation]**
2. A plot of paired observations used to examine association is called a _____ diagram. **[Scatter]**
3. Variables that tend to increase together have _____ correlation. **[Positive]**
4. Variables that tend to move in opposite directions have _____ correlation. **[Negative]**

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